Biomarker Optimization in Astronaut Digital Twins: A Multimodal Framework for Personalized Space Medicine

Kevin Dominey[1,2*][0009-0007-5374-1514], Cristian Vizitiu[1,3][0000-0003-2435-5028], and Ciprian Dobre[2][0000-0003-4638-7725]

[1] The Space Applications and Technologies Laboratory, Institute of Space Science – INFLPR Subsidiary, 077125 Măgurele, Romania
[2] National University of Science and Technology Politehnica Bucharest, 060042 Bucharest, Romania
[3] Department of Automatics and Information Technology, Faculty of Electrical Engineering and Computer Science, Transilvania University of Brasov, 500036 Brașov, Romania
kevin.dominey@stud.acs.upb.ro

Abstract. Long-duration spaceflight exposes astronauts to microgravity-induced physiological deconditioning under conditions of communication delay and scarce medical resources, demanding autonomous and personalized countermeasure strategies. Current practice selects countermeasure protocols from population-derived guidelines, resolving trade-offs between competing human performance and mission objectives implicitly rather than explicitly, and without formally characterizing what is physiologically attainable for a given individual. This paper presents a computational framework that reframes countermeasure selection as a joint optimization problem over two coupled structures. A *feasibility model*, formulated as a Dynamic Bayesian Network, projects the personalized, multivariate, and time-indexed probability distribution of biomarker configurations reachable from an astronaut's current state. A *desirability model*, overlaid on the same space, formalizes competing health, performance, and mission objectives as a Pareto structure anchored to an individualized reference configuration, penalizing deviations from that reference. Countermeasure selection is thereby reduced to identifying interventions that concentrate feasibility probability mass within desirable regions of the state space. Extending a prior Astronaut Human Digital Twin architecture, the framework specifies its functional layers, biomarker acquisition constraints, data integration requirements, and a descriptive–predictive–prescriptive analytical core. By distinguishing what is physiologically reachable from what is operationally desirable, it provides a formal basis for personalized countermeasure prescription in deep-space missions.

Keywords: Human Digital Twin · space medicine · precision healthcare · multi-objective optimization · Dynamic Bayesian Networks

1 Introduction

Long-duration spaceflight exposes astronauts to microgravity-induced physiological deconditioning, including bone loss, muscle atrophy, cardiovascular remodeling, and immune dysregulation, necessitating autonomous precision countermeasures amid the

communication delays and scarce medical resources characteristic of dynamic operational environments. Current space medicine relies on population-derived univariate biomarker reference intervals which ignore critical interdependencies and fail to balance competing objectives. There are studies that show inter-individual variability among astronauts, but to our knowledge no complete tool or framework currently integrates individual astronaut physiological data to compute physiologically feasible, multivariate biomarker targets tailored to mission-specific trade-offs.

Building on the prior Astronaut Human Digital Twin (HDT) framework definition developed by the research team [1], this paper proposes formal models separating what is physiologically reachable for an individual astronaut from what is operationally desirable, and specifies how the two are realized across the HDT's acquisition, standardization, analysis, and output components.

2 Background

Aerospace medicine is shifting from reactive, generalized countermeasures toward personalized, predictive precision healthcare [2], driven by the recognition of significant inter-individual variability in how astronauts adapt to stressors such as microgravity and radiation [3]. Traditional preselected, hypothesis-driven targets often fail to detect unanticipated "off-target" biological effects [4], whereas a holistic multimodal approach paired with machine learning can exploit high-dimensional biometric and omics data to characterize an individual's unique physiological fingerprint in response to the extreme spaceflight environment.

Identifying candidate biomarkers of resilience or vulnerability is a critical prerequisite for a robust precision space medicine system such as an Astronaut HDT. Because spaceflight affects nearly every biological system [5], a multi-domain biomarker library is required: for example, musculoskeletal resilience may be indexed by baseline myostatin and IGF-1 [3], cardiovascular deconditioning can be indexed by hemodynamics and heart rate variability [6, 7], and tissue stress can be indexed by cell-free DNA and RNA, whose spikes are associated with mitochondrial dysregulation [8]. Genomic and transcriptomic biomarkers also present hallmark signatures under spaceflight exposure, including telomere elongation and suppression of certain microRNAs [9–11]. There is also evidence linking gene expression patterns normally associated with aging-related frailty to the spaceflight environment [12].

Because spaceflight biometric and multi-omic data are rare and costly to collect, standardization and aggregation initiatives are improving their interoperability and accessibility. The *International Standards for Space Omics Processing* (ISSOP) consortium standardizes sample processing protocols and metadata normalization across agencies, and seeks to democratize access to public data repositories [13]. One such repository is the *Space Omics and Medical Atlas* (SOMA), which aggregates clinical, cellular, and multi-omic spaceflight data from cohorts including the NASA Twins Study, the JAXA CFE study, and the SpaceX Inspiration4 study [14–17], and constitutes a candidate source of flight data for informing population-level priors and retrospective validation of the proposed framework.

3 Multimodal Framework for Personalized Space Medicine

Optimization Models. Current spaceflight countermeasure practice faces two distinct shortcomings that a computational model is well-positioned to address. First, there is no systematic means of determining which physiological states are attainable for a given astronaut over a given mission interval, and second, trade-offs between competing human performance and mission objectives lack an explicit representation

that would enable their analysis and comparison. This paper proposes two coupled constructs, termed the feasibility model and the desirability model, to specifically address these aspects, whose general concept is illustrated in Fig. 1.

Feasibility model. For a given astronaut in a given mission context, the feasibility model defines the probability distribution over multivariate biomarker configurations reachable from their current state at a specified time in the future. It is proposed as a version of a Dynamic Bayesian Network (DBN) because it encodes the evolution of conditional dependencies between variables as a function of time. At any given moment, an individual's biomarker configuration is estimated with narrow uncertainty, but as the model projects forward the probability distribution shifts away from the current snapshot and spreads over a larger possibility space of configurations, each assigned a probability density given a specified $t + \Delta t$. This is illustrated by the nested contours A, B, and C in Fig. 1. The distribution is personalized (conditioned on individual physiology), multivariate (capturing joint constraints), and time-dependent (a larger change is more likely at a larger $t + \Delta t$). Importantly, changing countermeasure protocols reshapes the probability distribution. For instance, shifting a fixed allocation of onboard exercise time between resistive and aerobic modalities shifts the probability mass between domains, and concurrent training interference can make the reallocation not strictly linear. This makes the joint distribution, rather than independent per-biomarker projections, the appropriate object of analysis.

Desirability model. Overlaid on the same biomarker state space as the feasibility model, the desirability model maps biomarker configurations to a vector of competing human performance and mission objectives such as cardiovascular fitness and musculoskeletal preservation, evaluated under resource constraints such as crew time and caloric cost. Evaluating candidate protocols against this mapping yields a Pareto frontier of non-dominated objective vectors, making trade-offs explicit and analyzable. This is the support tool that will ultimately help mission planners make informed decisions regarding optimal individual countermeasure protocols. Fig. 1 represents this output as the green-to-red gradient background, with desirable configurations depicted as green and undesirable as red.

A defining characteristic of the human spaceflight countermeasure context is that the desirable region is anchored to an individualized reference configuration centered around the astronaut's pre-flight baseline. The realistic operational expectation is decline under microgravity exposure, making the primary objective to constrain that decline within acceptable bounds while ensuring a viable return-to-gravity state. The penalty structure around the reference is biomarker-specific: for measures where greater capacity carries no physiological cost (e.g., aerobic capacity) the objective is formulated as a one-sided penalty in which the baseline acts as a floor and improvements incur no cost, and for homeostatically regulated measures deviation in any direction is adverse and a two-sided penalty applies. The Pareto frontier is therefore defined over competing deviation costs under a fixed resource budget.

The clinically actionable query operates at the intersection of the feasibility and desirability models: for a given countermeasure protocol and time horizon, how much feasibility probability mass falls within the desirable regions of the state space? Fig. 1 depicts this: the feasibility mass drifts towards configurations that lie outside the most desirable region, so protocol modification (e.g., greater resistance-training allocation, nutritional or pharmacological support) is needed to redistribute mass toward it. Simulating alternative protocols across multiple time horizons produces a family of such feasibility-desirability landscapes, with different crewmembers receiving different prescriptions according to their individual vulnerabilities. The proposed framework thus reduces countermeasure protocol selection to choosing interventions whose projected landscape concentrates the most feasibility probability in the most desirable regions at each mission-critical timepoint.

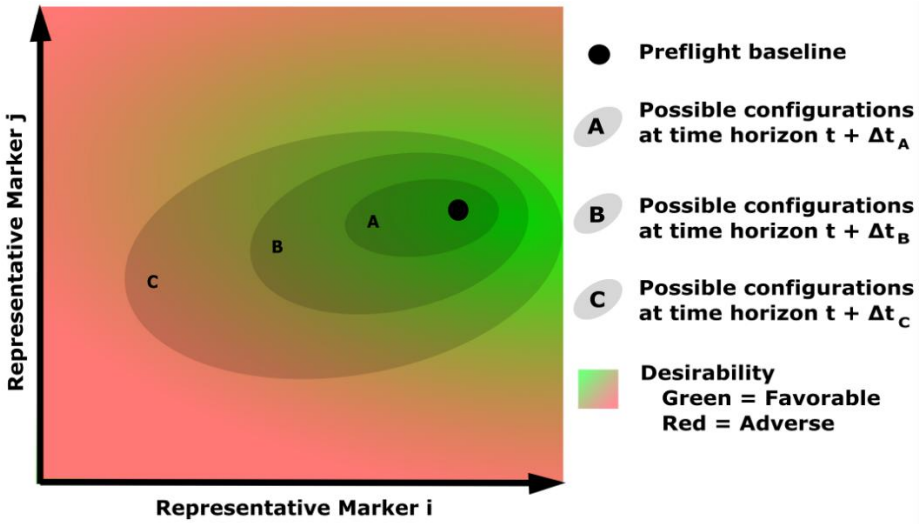


Fig. 1. Conceptual illustration of the feasibility and desirability models over a two-marker state space. Axes and values are schematic, not computed.

Framework Components. The framework operates across three functional layers: data collection, analysis & modeling, and decision support. These are each applied to three data categories: individual baseline, real-time, and predictive [1]. Baseline data (the pre-flight profile) serve a dual role: they provide the individual-level priors that differentiate crewmembers' initial feasibility models, and define the reference configuration against which desirability is evaluated. Real-time data, acquired through wearables, imaging, and periodic assessments, continuously update the HDT and validate prior predictions. Predictive data (i.e., projected biomarker distributions and associated risk trajectories) feed forward into decision support and are later validated against future real-time data for model retraining. This section describes how the feasibility and desirability constructs are realized within the data pipeline of the existing HDT architecture definition, presented here as the four architectural framework components FC1-FC4: FC1 and FC2 realize the data collection layer, FC3 the analysis and modeling layer, and FC4 the decision support layer.

FC1 Biomarker Acquisition. FC1 must deliver data usable by the feasibility model, which constrains panel composition through four requirements. *Domain coverage*: the panel must span the physiological systems most affected by spaceflight, including musculoskeletal, cardiovascular, metabolic, endocrine, and autonomic domains. Domain omission risks becoming unexplained variance in the model's probability shifts. *Measurability*: biomarkers must be obtainable through point-of-care assays, wearables, or minimally invasive sampling compatible with onboard capabilities. *Sensitivity*: biomarkers must be responsive to protocol changes within operationally relevant timeframes. *Inter-domain coupling*: biomarkers are selected not only for individual informativeness but also for the physiological dependencies they reveal when monitored jointly, since the model's value lies in the ability to analyze coupled systems.

FC2 Standardization and Storage. FC2 transforms the heterogeneous acquisition data into the representation that the model consumes: temporal alignment reconciles continuous wearable streams with biochemical assays or fitness tests obtained days or weeks apart. Normalization across units and scales establishes comparability, and absolute biomarker values are recast as deviations from reference configurations. The Bayesian structure tolerates missing data, since individual posterior updates may be skipped without breaking the model, though repeated omissions compound uncertainty over the projection horizon. Onboard storage and compute constraints motivate

prioritization of edge processing for feature extraction and dimensionality reduction, with raw measurements retained locally only as long as needed for prediction validation.

FC3 Analytical Core. FC3 is where the feasibility and desirability models reside. Its three stages are descriptive, predictive, and prescriptive. The descriptive stage characterizes the astronaut's current position in multivariate biomarker space with uncertainty bounds, performing anomaly detection, trend identification, and cross-domain correlation. The predictive stage instantiates the feasibility model: given the current state estimate, active countermeasure protocol, and a time horizon, the Bayesian state-transition model generates a probability distribution over future biomarker configurations. Initialized with population-level priors from spaceflight, analog, and mechanistic sources, it individualizes progressively as observations accumulate. The prescriptive stage instantiates the desirability model, applying multi-objective optimization to yield a Pareto frontier across competing objectives. Because onboard biomarker sampling is intermittent and noisy, the astronaut's true physiological state is never directly observed; countermeasure selection therefore operates over a belief distribution rather than a known state, making the decision layer a type of Partially Observable Markov Decision Process.

FC4 Outputs and Interpretability. FC4's primary output is the Pareto frontier of non-dominated objective vectors produced by FC3 for a given time horizon, each point corresponding to a candidate countermeasure protocol and its predicted outcomes. Because each point corresponds to specific predicted physiological values, the frontier itself also functions as the framework's main interpretability device: movement along it is a quantifiable trade-off between deviation costs, letting mission planners prioritize by mission stage while retaining insight and agency over the decision. Thus, interpretability is designed into the framework, offsetting the opacity of the machine learning components of the analytical core. Secondary outputs include risk alerts, anomaly detection, and the trained model itself for use with future crews.

4 Conclusions and Future Work

This paper has presented a computational framework that reframes astronaut countermeasure selection as a joint optimization problem over two coupled structures: a feasibility model projecting the personalized, multivariate, time-indexed probability distribution over reachable biomarker configurations, and a desirability model formalizing competing health, performance, and mission objectives as a Pareto structure anchored to an individualized reference configuration. By distinguishing what is physiologically attainable from what is operationally desirable, the framework moves beyond monitoring and anomaly detection toward explicit, individualized countermeasure prescription.

The framework is presented at the architectural level, and several components remain to be formalized. The algorithms populating the feasibility distribution are not yet fixed; the planned proof-of-concept will use particle-based Bayesian state estimation for the feasibility model and biomarker-specific deviation penalty functions for the desirability model, and future work will specify these internals along with the belief-state representation and update mechanism underlying the decision layer. Validation will proceed along a staged roadmap, from synthetic data to retrospective spaceflight and analog cohorts, followed by prospective ground-based evaluation and ultimately in-flight deployment. Further work will address objective-function composition through dimensionality reduction, extension of the biomarker panel across additional domains, and integration of this framework into onboard autonomous spaceflight medical systems.

Acknowledgments. This work was supported by the PubArt programme of the National University of Science and Technology POLITEHNICA Bucharest, and by the following projects: Digital4Sustainability, ERASMUS-EDU-2023-PI-ALL-INNO, contract no. 101140316/25.01.2024, CRESCDI code 020235498; GreenShift – EU MSc in AI & HPC for Green Digital Innovation in Transport, contract no. 101226222/30 June 2025, CRESCDI code 020256765; Support for the Development and Strengthening of DANUBIUS-RO (DANUBIUS-RO-2), PN IV, Subprogramme 5.9.4, contract no. 1D of 03.10.2025, signed with the National Authority for Research; and Empowering the Future through Innovative Smart Solutions for Rural Areas (FUTURAL), contract no. 101083958, CRESCDI code 020234823.

References

1. Dominey, K., Nistorescu, A., Marin, M., Vizitiu, C., Dobre, C.: Human Digital Twin for Astronauts: Integrating Microgravity Stress Testing and AI-Driven Predictive Modeling for Human Health in Space. In: Advances in Digital Health and Medical Bioengineering II. EHB 2025. IFMBE Proceedings, vol. 144. Springer, Cham (2026). doi: 10.1007/978-3-032-23952-5_2
2. Schmidt, M.A., et al.: Why Personalized Medicine Is the Frontier of Medicine and Performance for Humans in Space. New Space **8**, 63–76 (2020). doi: 10.1089/space.2019.0037
3. Minoretti, P., et al.: Potential Biomarkers of Resilience to Microgravity Hazards in Astronauts. Cureus **16**, e57173 (2024). doi: 10.7759/cureus.57173
4. Schmidt, M.A., Goodwin, T.J.: Personalized medicine in human space flight: using Omics based analyses to develop individualized countermeasures that enhance astronaut safety and performance. Metabolomics **9**, 1134–1156 (2013). doi: 10.1007/s11306-013-0556-3
5. DeVirgiliis, L., et al.: Spaceflight and sport science: Physiological monitoring and countermeasures for the astronaut–athlete on Mars exploration missions. Exp. Physiol. (2025). doi: 10.1113/EP091595
6. Gallo, C., et al.: Cardiovascular deconditioning during long-term spaceflight through multiscale modeling. npj Microgravity **6**, 27 (2020). doi: 10.1038/s41526-020-00117-5
7. Jones, C.W., et al.: Molecular and physiological changes in the SpaceX Inspiration4 civilian crew. Nature **632**, 1155–1164 (2024). doi: 10.1038/s41586-024-07648-x
8. Husna, N., et al.: Release of CD36-associated cell-free mitochondrial DNA and RNA as a hallmark of space environment response. Nat. Commun. **15**, 4814 (2024). doi: 10.1038/s41467-023-41995-z
9. Luxton, J.J., et al.: Temporal Telomere and DNA Damage Responses in the Space Radiation Environment. Cell Rep. **33**, 108435 (2020). doi: 10.1016/j.celrep.2020.108435
10. Malkani, S., et al.: Circulating miRNA Spaceflight Signature Reveals Targets for Countermeasure Development. Cell Rep. **33**, 108448 (2020). doi: 10.1016/j.celrep.2020.108448
11. Garrett-Bakelman, F.E., et al.: The NASA Twins Study: A multidimensional analysis of a year-long human spaceflight. Science **364**, eaau8650 (2019). doi: 10.1126/science.aau8650
12. Camera, A., et al.: Aging and putative frailty biomarkers are altered by spaceflight. Sci. Rep. **14**, 13098 (2024). doi: 10.1038/s41598-024-57948-5
13. Rutter, L., et al.: A New Era for Space Life Science: International Standards for Space Omics Processing. Patterns **1**, 100148 (2020). doi: 10.1016/j.patter.2020.100148
14. Overbey, E.G., et al.: The Space Omics and Medical Atlas (SOMA) and international astronaut biobank. Nature **632**, 1145–1154 (2024). doi: 10.1038/s41586-024-07639-y
15. Overbey, E.G., et al.: Collection of biospecimens from the Inspiration4 mission establishes the standards for the Space Omics and Medical Atlas (SOMA). Nat. Commun. **15**, 4964 (2024). doi: 10.1038/s41467-024-48806-z
16. Muratani, M.: Cell-free RNA analysis of plasma samples collected from six astronauts in JAXA Cell-Free Epigenome (CFE) Study. NASA Open Science Data Repository, Version 3 (2022). doi: 10.26030/r2xr-h714
17. Kim, J., et al.: Single-cell multi-ome and immune profiles of the Inspiration4 crew reveal conserved, cell-type, and sex-specific responses to spaceflight. Nat. Commun. **15**, 4954 (2024). doi: 10.1038/s41467-024-49211-2